

Gustavo Fróes do Vale

gustavo.froesdovale@telecom-paris.fr

Education

Institut Polytechnique de Paris - Télécom Paris , Double Degree Exchange Program, B.S. in Engineering (M1) and MSc in Engineering (M2)	Sep 2024 – Present Paris, France
• M1 - Study Tracks: Applied Algebra: Cryptography, Quantum information, Coding theory; MODS: Markets, Organizations, Data, Strategies. GPA: 3.8/4.0 • M2 - QMI: Quantum, Mathematics, and Computer Science Key Courses: Quantum non-locality, Convex Optimization, Quantum Shannon theory, Quantum structures	
University of Campinas (UNICAMP) , B.S. in Physics	May 2021 – Present Campinas, Brazil
• Key Courses: Quantum Mechanics (I and II), Statistical Mechanics, Classical Mechanics (I and II), Advanced Linear Algebra. GPA: 0.88/1.0	

Undergraduate Research Experience

Curiosity Team (QTY) - Summer internship - Télécom Paris, France <i>Search for counterexamples of the revised Peres conjecture</i>	Jun 2025 – Aug 2025
• Learned numerical methods for lower-bounding distillable key rates in device-independent quantum key distribution (DIQKD) protocols. • Conducted a search for bound entangled states capable of producing a positive key rate in DIQKD. Research Advisor: Prof. Peter J. Brown Télécom Paris	
Mathematical Foundations of Quantum Theory (MFQ) - UNICAMP, Brazil <i>Maximal Violations of Noncontextual Inequalities</i>	Sep 2023 – Aug 2024
• Studied the formalism behind Bell's non-locality and Kochen-Specker contextuality. • Developed a numerical software tool to calculate the maximum violation of contextual inequalities, such as the KCBS inequality, inspired by the See-Saw algorithm and using the optimization software Gurobi. Research Advisor: Prof. Rafael Rabelo IFGW - UNICAMP	
Scientific initiation Program - PIBIC - UNICAMP, Brazil Funding: CNPq <i>Sensitivity Study of the DUNE Experiment for the Detection of Sterile Neutrinos</i>	Sep 2022 – Sep 2023
• Conducted a detailed analysis of the sensitivity of the DUNE (Deep Underground Neutrino Experiment) in detecting a possible new type of particle (Sterile Neutrino), focusing on different regions of the parameter space using the software GLoBES. • Concluded that DUNE will be capable of excluding the sterile neutrino hypothesis with 95% confidence, while imposing limits on new oscillation parameters in scenarios with a 3+1 neutrino model. Research Advisor: Prof. Pedro Cunha de Holanda IFGW - UNICAMP	
Scientific initiation and Master's Program - PICME - UNICAMP, Brazil Funding: CNPq <i>Introduction to Stochastic Differential Equations and Simulations - 6 Months</i>	Aug 2021 – Aug 2022
• Gained foundational knowledge of stochastic differential equations (SDEs) and their application to financial modeling. • Explored and implemented key models for asset price prediction, including the Black-Scholes model for option pricing. • Developed simulations to solve and analyze SDEs, using computational tools to visualize and validate results. Research Advisor: Prof. Pedro Jose Catuogno IMECC - UNICAMP	

Angular Deviation From Snell's Law in Gaussian beams - 6 Months

- Conducted a study on the angular deviations predicted by Snell's law and analyzed the effects of transverse symmetry breaking in laser beam propagation through dielectric blocks.
- Performed comparative analysis between theoretical approximations and real-world beam data, validating the accuracy of approximations.

Research Advisor: Prof. Stefano De Leo | IMECC - UNICAMP

Teaching

Tutor in Classical Mechanics I - UNICAMP, Brazil

Aug 2023 – Dec 2023

- Conducted one-on-one tutoring sessions to assist students in grasping complex concepts and improving academic performance
- Demonstrated strong communication and interpersonal skills to build rapport with students and foster a positive learning environment

Extra Curricular

Quantum Coaches (UNICAMP's International Physicists' Tournament team) -

Aug 2021 – Apr 2023

UNICAMP, Brazil

Core member

- Gained experience in real research in physics by solving several open-ended problems
- Participated in both national and international competitions, winning several awards
- Attained valuable public speaking skills, winning an award for it

Supervisor: Prof. Pierre Louis de Assis | IFGW - UNICAMP

Honors & Awards

International Honors

9th Place, 15th International Physicists' Tournament (IPT)

2023 – Paris, France

Best Reporter Award, 15th International Physicists' Tournament (IPT)

2023 – Paris, France

2nd Place, 14th International Physicists' Tournament (IPT)

2022 – Bucaramanga,
Colombia

Domestic Honors

1st Place, 5th Brazilian Physicists' Tournament (BPT)

2022 – Campinas, Brazil

1st Place, 4th Brazilian Physicists' Tournament (BPT)

2022 – Online

Bronze Medal, Brazilian Physics Olympiad (OBF)

2019 – Brasília, Brazil

Bronze Medal, Brazilian Mathematics Olympiad (OBMEP)

2019 – Brasília, Brazil

Silver Medal, Brazilian Astronomy Olympiad (OBA)

2017 – Brasília, Brazil

Skills

Programming Languages: Python, C++ , Wolfram Mathematica

Technologies: Gurobi, GLOBES, LaTeX, Pandas, Numpy, Ncpol2sdpa, Matplotlib, Cvxpy

Languages: Portuguese (Native), English (Advanced), French (Basic/Intermediate)

References

Rafael Luiz da Silva Rabelo

University of Campinas, Physics Institute Gleb Wataghin
Rua Sérgio Buarque de Holanda 777
Campinas, SP - Brazil
Email: rrlabelo@unicamp.br

Peter J. Brown

Télécom Paris, QTY Team
19 Pl. Marguerite Perey
91120 Palaiseau - France
Email: peter.brown@telecom-paris.fr